

1. Questions

Read the following data carefully and answer the questions based on it.

A company manufactures five different types of articles i.e. A, B, C, D and E. Weight and Cost price of each article is different. The given table shows the number of each type of article produced by the company in January, cost price of the material used in producing the article and weight of each type of article.

Article	Number of articles produced	Cost of material (in Rs/kg)	Weight of an article(in kg)
A	$(s/q)^2 + 7$	$(q^2 + p^{1/2})$	$(r/q)^2$
B	$p^2 - 2q$	$[(s+3)/3] + 1$	$2(s-5)$
C	$2r$	$r/2$	$2p$
D	$3p - s/7$	$2p+2$	$3r + 3$
E	$2s$	r	$(s+p)*133.33\%$

Notes:

- i). Cost price of article = Total cost price of material (Rs.) + Making Charge
- ii). Total cost price of material (Rs.)= Cost of material (in Rs/kg) * Weight of an article (in kg)
- iii). Number of articles produced = Number of articles sold + number of articles unsold
- iv). Making charge for each article is 16.66% of the cost price of the total material used in producing it. The cost price of article C is 22.5% of the cost price of article E. The company earned a profit of 12.5% when it sold article D for Rs 1260. The weight of article A is 25 kg, and the number of article C produced by the company is 6 less than 1.5 times the number of article D produced.

A revenue of Rs 18900 is generated by selling “M%” of article B. If after paying a tax of 10%, the net profit of the company is 12.5%, then find the value of M?

- a. 40
- b. 45
- c. 60
- d. 65
- e. None of these

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2. Questions

Out of the total number of article A produced by the company, 14.28% were defective and sold at a loss of 16%. 33.33% of the remaining articles were sold at a profit of 12%, and the remaining were sold at a profit of 10%. Find the profit or loss incurred on selling all the article A produced by the company?

- a. Rs 1526

- b. Rs 1424
- c. Rs 1284
- d. Rs 1344
- e. None of these

3. Questions

Find the difference between the total cost price of material used in producing total number of article D and E?

- a. $r^3 - 135p$
- b. $10ps + 280$
- c. $3s^2 + 835$
- d. Both A and B
- e. Both A and C

4. Questions

The number of article X produced by the company is 33.33% more than that of article C and cost price of article X is 66.66% more than that of B. If the making charge on producing article X is 12%, then find the cost price of total material used in producing total number of article X?

- a. Rs 18000
- b. Rs 20000
- c. Rs 25000
- d. Rs 21000
- e. None of these

5. Questions

The average of the cost price of article A, article B, and article E together is what percent of the cost price of article D?

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- a. 46.25%
- b. 35.25%
- c. 40.25%
- d. 41.25%
- e. None of these

6. Questions

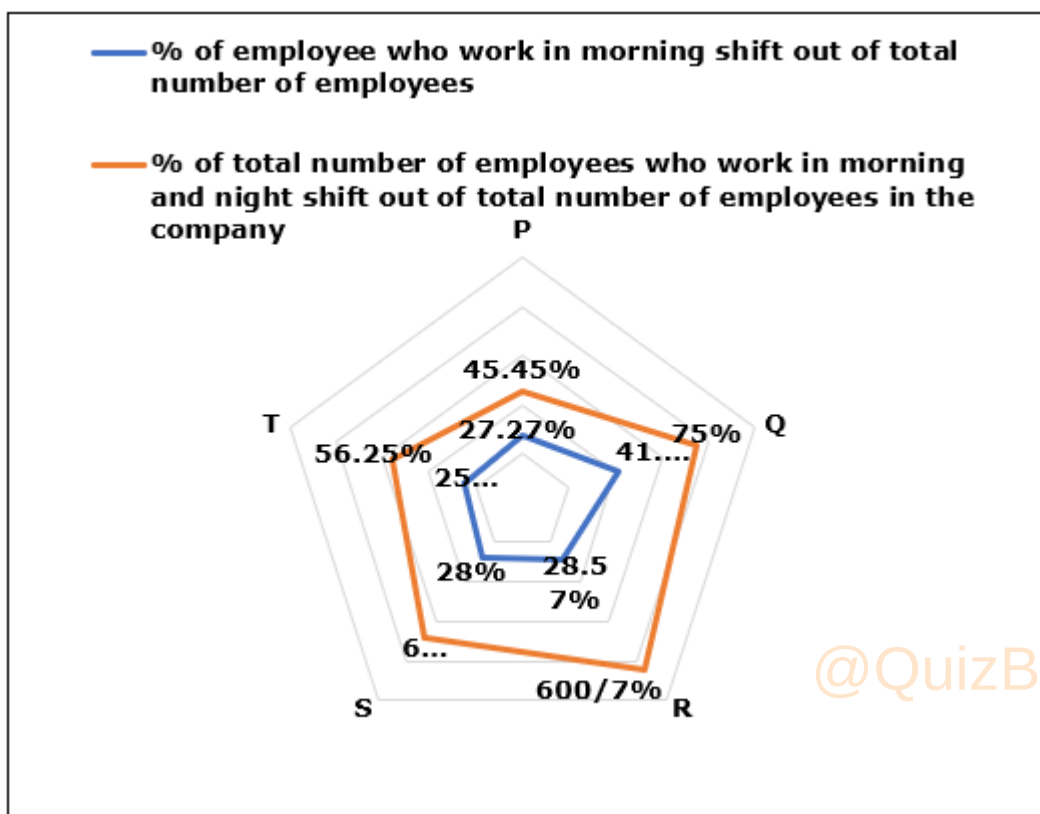
If the cost price of material used in producing an article C increase by 60%, then by how much percent the weight of an article should be reduced such that there will be no change in the total cost price of material (Rs.)?

- a. 22.5%
- b. 33.5%
- c. 37.5%
- d. 27.5%
- e. None of these

7. Questions

Read the following data carefully and answer the questions based on it.

There are five companies (P, Q, R, S, and T) and different numbers of employees working in three different shifts (morning, night, and evening shift). The given radar graph shows the percentage of number of employees work in morning shift in each company and also shows the percentage of total number of employees who work in morning and night shift together out of total number of employees in the company. No company has more than 1000 employees.



The difference between the number of employees who work in the evening shift of company Q and R is 34, and the ratio of the total number of employees who work in company P and Q is 22:19, respectively. The number of employees who work in company R is 10.4% less than that of company S and the number of employees who work in company Q is the lowest among all these five companies. The number of employees who work in the evening shift in company T is 210, which is equal to the sum of the number of employees who work the night shift in company P and the number of employees who work the evening

shift in company Q.

If the difference between the number of employees who work in the evening and night shift in company Q is “M” and the difference between the number of employees who work in the morning and night shift of company S is “N”, then find the value of “M+N”?

- a. 113
- b. 125
- c. 108
- d. 117
- e. None of these

8. Questions

Two quantities are given below. Solve both of them and establish the relationship between Quantity I and Quantity II.

Quantity I: Find the average number of employees who work in morning shift in companies R and T.

Quantity II: Find the average number of employees who work in night shift in companies Q and T.

- a. $Q(I) \times 2.5 = Q(II) \times 2 + 46$
- b. $Q(I) > [Q(II) - 10]$
- c. $[Q(I) + 15] > [Q(II) + 2]$
- d. Both A and B
- e. Both B and C

9. Questions

Out of the total number of employees who work in company Q, 33.33% are females, and the number of males who work in night shift in company Q is three times that of females. Find the difference between the number of males and females who work in the morning and night shift together in company Q?

- a. 84
- b. 57
- c. 71
- d. 76
- e. None of these

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10. Questions

If the number of employees in company A is 20% less than that in company S and 25% and 10% of employees work in the morning and night shift, respectively, then the number of employees

working in the evening shift in company A is how much more than that in company P?

- a. 29
- b. 37
- c. 32
- d. 41
- e. none of these

11. Questions

The average number of employees working in night shift in all companies together is $(W+16.66)\%$ of the total number of employees in company P. Find $W\%$ of the average number of employees in companies R and T together.

- a. 98
- b. 112
- c. 104
- d. 96
- e. None of these

12. Questions

In each question, a number series is given. Read all the questions carefully and answer the questions based on it.

Two series are given below; both contain a missing term.

I: $[3B - 2]$, $[2(A+B) + 14]$, $[B^2 + 23]$, ?, $[3(A-B)^3 + 141]$, $[839 \cdot (B)^{0.5}]$

II: 78, $[6A + 15]$, ?, $[9(A+2B)+24]$, 852, $[(B-5)^3 - 38]$

If A and B are the positive roots of the equation $x^2 - 40x + 384 = 0$ and A is greater than B, then find the difference between the missing terms of both the series?

- a. 263
- b. 235
- c. 247
- d. 257
- e. None of these

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13. Questions

A series is given: 12, 21, 30, 43, 60, 79. It contains a wrong term, and the term which will replace the odd term in the series is "P". If another series starts with "7P" and follows the same pattern and

the 6th term of the new series is “Y”, then 2P is what percent of Y?

- a. 9%
- b. 19%
- c. 38%
- d. p%
- e. Both (b) and (d)

14. Questions

Three series (I), (II), and (III) are given below. Each series contains one wrong term and wrong in series I, II, and III is denoted by X, Y and Z respectively.

I: 336, 348, 364, 380, 408, 436

II: 421, 457, 506, 580, 690, 849

III: 56, 364, 1820, 6370, 12740, 6375

Find the value of $[(Z-Y)/2] \div [(25\% \text{ of } X)/5+7]$?

- a. 120.5
- b. 136.5
- c. 114.5
- d. 128.5
- e. None of these

15. Questions

Two series are given below.

I: 103, 128, 249, 538, (P+Q), 2028

II: P, R, 436, 315, 484, 259, 548

Establish the relationship between 1.8R and Q.

- a. $1.8R < Q$
- b. $1.8R > Q$
- c. $1.8R = Q$
- d. $1.8R \leq Q$
- e. $1.8R \geq Q$

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16. Questions

Vihan borrowed Rs (12A+10680) from the bank for 3 years at a 12.5% rate of interest

compounded annually. He lent out 60% of this amount to Shruti at a 15% simple rate of interest and the remaining to Prachi at a 25% rate of interest compounded annually for three years. At the end of three years, Vihan earned a profit of Rs 10479.6.

Which among the following statement is true?

I: The value of A is divisible by 11.

II: When A is invested for 2 years at 12% rate of simple interest, the amount obtained at the end of two years is 3648.

III: The value of A is 75 less than a perfect square.

- a. Only I
- b. Only II
- c. Only III
- d. Both II and III
- e. None of these

17. Questions

There are three boats P, Q and R sail on the river which is flowing with the speed of $(X - 7)$ kmph. Boat P can cover 156 km upstream distance in (X) hours which is 62.5% more than the time taken by boat Q to cover (Y) km downstream distance. The difference between upstream speed of boat R and downstream speed of boat P is 9 kmph. Downstream distance covered by boat P in (Z) hours is 60% more than the upstream distance covered by boat R in same time.

From the statement given in the question which among the following can be determined?

- a. Value of Y
- b. Value of X
- c. Value of Z
- d. Both Value of X and Z
- e. None of these

18. Questions

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There are four members (A, B, C, and D) in the family. The age of C after 6 years is a perfect square as well as a perfect cube (C's age is a two digit number). The ages of A is 8 years older than age of D is 8 years. The ratio of the present age of A and age of B after 9 years is 7:6. The sum of the ages of A and C before 6 years is 14.28% more than the sum of the ages of B and D after 8 years. Find the sum of the present age of B and C.

- a. 72 years
- b. 69 years
- c. 63 years

d. 60 years

e. 85 years

19. Questions

In a three-digit number, the digit at the 100th place is the sum of the digits at the unit and tens place. When we reverse the digits of the number, it becomes 297 less. If all the digits are distinct, then which among the following can be determined?

- a. Sum of digits at tens and unit place
- b. The digit at the tens place
- c. All three digits of number
- d. Difference between digits at the tens and unit place
- e. None of these

20. Questions

M can finish a piece of work in (Y-4) days, whereas the time taken by O to finish the same work is (Y+5) days. The ratio of efficiency of N and O is 6:7, respectively. M and N together can finish the work in $13\frac{1}{8}$ days. Which among the following statements is/are true about the value of Y?

- a. The value of Y is a prime number
- b. Y is divisible by the sum of its digits
- c. The sum of digits of the square of Y is a prime number
- d. Both A and C
- e. None of these

Explanations:

1. Questions

The cost price of article C is 22.5% of the cost price of article E.

Cost price of article E = $40x$

Cost price of article C = 22.5% of $40x = 9x$

Weight of article C = $2p$ kg

Weight of article E = $(s+p)*\frac{4}{3}$ kg

Cost price of material used in C = Rs. $\frac{r}{2}$ per kg

Cost price of material used in E = Rs. r per kg

The number of article C produced by the company is 6 less than 1.5 times the number of article D

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produced.

$$(2p * (r/2) * (7/6))/[(s + p) * (4/3) * r * (7/6)] = 9x/40x$$

$$3p/[4 * (s + p)] = 9/40$$

$$10p = 3p + 3s$$

$$7p = 3s$$

$$\text{let } p = 3a \text{ and } s = 7a$$

$$\text{Weight of article A} = (r/q)^2 = 25 \text{ kg}$$

$$r/q = 5$$

$$\text{let } r = 5b \text{ and } q = b$$

The number of article C produced by company is 6 less than one and half times the number of article D produced

$$1.5 * (3p - s/7) - 6 = 2r$$

$$1.5(9a - a) - 6 = 10b$$

$$12a - 6 = 10b$$

$$6a - 3 = 5b \text{ ---(I)}$$

The company earned a profit of 12.5% when it sold article D for Rs 1260

$$\text{Cost price of article D} = 1260/9 * 8 = \text{Rs } 1120$$

$$(2p+2) * (3r+3) * 7/6 = 1120$$

$$(p+1)(r+1) = 160$$

$$(3a+1)(5b+1) = 160$$

$$5b + 1 = 160/(3a+1)$$

$$5b = 160/(3a+1) - 1 \text{ ---(II)}$$

From equations I and II

$$6a - 3 = 160/(3a+1) - 1$$

$$6a - 2 = 160/(3a+1)$$

$$(3a-1)(3a+1) = 80$$

$$(3a)^2 - 1^2 = 80$$

$$9a^2 = 81$$

$$a = 3$$

$$b = 3$$

$$p = 9$$

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$$s = 21$$

$$q = 3$$

$$r = 15$$

Article	Number of articles produced	Cost price of material (Rs/kg)	Weight of an article (in kg)	Cost price of an article (in Rs)
A	56	12	25	350
B	75	9	32	336
C	30	7.5	18	157.5
D	24	20	48	1120
E	42	15	40	700

Answer: C

According to the questions,

Total selling price = Rs 18900

Tax paid = Rs 1890

Net revenue = 18900 – 1890 =Rs 17010

Cost price of article sold = $17010/9 \times 8$ =Rs 15120

Cost price of one article B = Rs 336

Number of article sold = $15120/336$ =45

Total number of article B produced = 75

$$45/75 \times 100 = M$$

$$M = 60$$

Hence, the answer is option C.

2. Questions

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Cost price of article E = $40x$

Cost price of article C = 22.5% of $40x = 9x$

Weight of article C = $2p$ kg

Weight of article E = $(s+p) \times 4/3$ kg

Cost price of material used in C =Rs. $r/2$ per kg

Cost price of material used in E =Rs. r per kg

The number of article C produced by the company is 6 less than 1.5 times the number of article D produced.

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$$(2p * (r/2) * (7/6))/[(s + p) * (4/3) * r * (7/6)] = 9x/40x$$

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$$10p = 3p + 3s$$

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E	42	15	40	700

Answer: D

Total number of article A produced = 56

Cost price of each article A = Rs 350

Number of defective articles = $56 \times \frac{1}{7} = 8$

Selling price of defective article = $350 \times \frac{84}{100} = 294$

Total selling price of defective articles = Rs 2352

Number of articles sold at 12% profit = $\frac{48}{3} = 16$

Selling price = $16 \times 350 \times 112\% = \text{Rs } 6272$

Number of articles sold at 10% profit = 32

Selling price = Rs 12320

Total Cost price of producing all article A = $350 \times 56 = \text{Rs } 19600$

Total Selling price = $2352 + 6272 + 12320 = \text{Rs } 20944$

Profit = $20944 - 19600$

= Rs 1344

Hence, the answer is option D.

3. Questions

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The cost price of article C is 22.5% of the cost price of article E.

Cost price of article E = $40x$

Cost price of article C = 22.5% of $40x = 9x$

Weight of article C = $2p$ kg

Weight of article E = $(s+p) \times \frac{4}{3}$ kg

Cost price of material used in C = Rs. $\frac{r}{2}$ per kg

Cost price of material used in E = Rs. r per kg

The number of article C produced by the company is 6 less than 1.5 times the number of article D produced.

$$(2p * (r/2) * (7/6)) / [(s + p) * (4/3) * r * (7/6)] = 9x/40x$$

$$3p / [4 * (s + p)] = 9/40$$

$$10p = 3p + 3s$$

$$7p = 3s$$

$$\text{let } p = 3a \text{ and } s = 7a$$

$$\text{Weight of article A} = (r/q)^2 = 25 \text{ kg}$$

$$r/q = 5$$

$$\text{let } r = 5b \text{ and } q = b$$

The number of article C produced by company is 6 less than one and half times the number of article D produced

$$1.5 * (3p - s/7) - 6 = 2r$$

$$1.5(9a - a) - 6 = 10b$$

$$12a - 6 = 10b$$

$$6a - 3 = 5b \text{ ---(I)}$$

The company earned a profit of 12.5% when it sold article D for Rs 1260

$$\text{Cost price of article D} = 1260/9 * 8 = \text{Rs } 1120$$

$$(2p+2) * (3r+3) * 7/6 = 1120$$

$$(p+1)(r+1) = 160$$

$$(3a+1)(5b+1) = 160$$

$$5b + 1 = 160/(3a+1)$$

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From equations I and II

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$$b = 3$$

$$p = 9$$

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A	56	12	25	350
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Answer: A

Cost price of material used in article D = $48 \times 20 \times 24 = \text{Rs } 23040$

Cost price of material used in article E = $42 \times 15 \times 40 = \text{Rs } 25200$

Required difference = $25200 - 23040$

$$= 2160$$

$$= r^3 - 135p$$

Hence, the answer is option A.

4. Questions

The cost price of article C is 22.5% of the cost price of article E.

Cost price of article E = $40x$

Cost price of article C = 22.5% of $40x = 9x$

Weight of article C = $2p$ kg

Weight of article E = $(s+p) \times \frac{4}{3}$ kg

Cost price of material used in C = Rs. $\frac{r}{2}$ per kg

Cost price of material used in E = Rs. r per kg

The number of article C produced by the company is 6 less than 1.5 times the number of article D produced.

$$(2p \times \frac{r}{2} \times \frac{7}{6}) / [(s+p) \times \frac{4}{3} \times r \times \frac{7}{6}] = 9x/40x$$

$$3p/[4 \times (s+p)] = 9/40$$

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$$10p = 3p + 3s$$

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$$\text{let } p = 3a \text{ and } s = 7a$$

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The number of article C produced by company is 6 less than one and half times the number of article D produced

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$$6a - 3 = 5b \text{ ---(I)}$$

The company earned a profit of 12.5% when it sold article D for Rs 1260

$$\text{Cost price of article D} = 1260/9 \times 8 = \text{Rs } 1120$$

$$(2p+2)(3r+3) \times 7/6 = 1120$$

$$(p+1)(r+1) = 160$$

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Answer: B

Total number of article X produced = $30 \times \frac{4}{3} = 40$

Cost price of producing an article X = $336 \times \frac{5}{3} = 560$

Making charges = 12%

Let the cost price of material = Rs 100m /kg

Cost price of material = $100m + (100m \times 12)\% = 560$

$112m = 560$

$m = 5$

Cost price of material = 500

Total cost price of material used = $500 \times 40 = \text{Rs } 20000$

Hence, the answer is option B.

5. Questions

The cost price of article C is 22.5% of the cost price of article E.

Cost price of article E = $40x$

Cost price of article C = 22.5% of $40x = 9x$

Weight of article C = $2p$ kg

Weight of article E = $(s+p) \times \frac{4}{3}$ kg

Cost price of material used in C = Rs. $\frac{r}{2}$ per kg

Cost price of material used in E = Rs. r per kg

The number of article C produced by the company is 6 less than 1.5 times the number of article D produced.

$$(2p \times (\frac{r}{2}) \times (\frac{7}{6})) / [(s + p) \times (\frac{4}{3}) \times r \times (\frac{7}{6})] = 9x/40x$$

$$3p/[4 \times (s + p)] = 9/40$$

$$10p = 3p + 3s$$

$$7p = 3s$$

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let $p = 3a$ and $s = 7a$

Weight of article A = $(r/q)^2 = 25$ kg

$$r/q = 5$$

let $r = 5b$ and $q = b$

The number of article C produced by company is 6 less than one and half times the number of article D produced

$$1.5*(3p - s/7) - 6 = 2r$$

$$1.5(9a - a) - 6 = 10b$$

$$12a - 6 = 10b$$

$$6a - 3 = 5b \text{ ---(I)}$$

The company earned a profit of 12.5% when it sold article D for Rs 1260

Cost price of article D = $1260/9*8 = \text{Rs } 1120$

$$(2p+2)*(3r+3)*7/6 = 1120$$

$$(p+1)(r+1) = 160$$

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E	42	15	40	700

Answer: D

Average of cost price of producing articles A, B and E = $(350+336+700)/3 = \text{Rs } 462$

Cost price of producing an article D = Rs 1120

Thus, percentage = $462/1120 \times 100$

=41.25%

Hence, the answer is option D.

6. Questions

The cost price of article C is 22.5% of the cost price of article E.

Cost price of article E = $40x$

Cost price of article C = 22.5% of $40x = 9x$

Weight of article C = $2p$ kg

Weight of article E = $(s+p) \times 4/3$ kg

Cost price of material used in C = Rs. $r/2$ per kg

Cost price of material used in E = Rs. r per kg

The number of article C produced by the company is 6 less than 1.5 times the number of article D produced.

$$(2p \times (r/2) \times (7/6)) / [(s + p) \times (4/3) \times r \times (7/6)] = 9x/40x$$

$$3p/[4 \times (s + p)] = 9/40$$

$$10p = 3p + 3s$$

$$7p = 3s$$

$$\text{let } p = 3a \text{ and } s = 7a$$

$$\text{Weight of article A} = (r/q)^2 = 25 \text{ kg}$$

$$r/q = 5$$

$$\text{let } r = 5b \text{ and } q = b$$

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The number of article C produced by company is 6 less than one and half times the number of article D produced

$$1.5*(3p - s/7) - 6 = 2r$$

$$1.5(9a - a) - 6 = 10b$$

$$12a - 6 = 10b$$

$$6a - 3 = 5b \text{ ---(I)}$$

The company earned a profit of 12.5% when it sold article D for Rs 1260

$$\text{Cost price of article D} = 1260/9*8 = \text{Rs } 1120$$

$$(2p+2)*(3r+3)*7/6 = 1120$$

$$(p+1)(r+1) = 160$$

$$(3a+1)(5b+1) = 160$$

$$5b + 1 = 160/(3a+1)$$

$$5b = 160/(3a+1) - 1 \text{ ---(II)}$$

From equations I and II

$$6a - 3 = 160/(3a+1) - 1$$

$$6a - 2 = 160/(3a+1)$$

$$(3a-1)(3a+1) = 80$$

$$(3a)^2 - 1^2 = 80$$

$$9a^2 = 81$$

$$a = 3$$

$$b = 3$$

$$p = 9$$

$$s = 21$$

$$q = 3$$

$$r = 15$$

@QuizByMickey

Article	Number of articles produced	Cost price of material (Rs/kg)	Weight of an article (in kg)	Cost price of an article (in Rs)
A	56	12	25	350
B	75	9	32	336
C	30	7.5	18	157.5
D	24	20	48	1120
E	42	15	40	700

Answer: C

Cost price of material used in producing article C = Rs 7.5/kg

New cost price = $7.5 \times 8/5 = \text{Rs } 12/\text{kg}$

Original weight of an article C = 18 kg

Cost price = $7.5 \times 18 = \text{Rs } 135$

weight if the cost price is increased = $135/12 = 11.25 \text{ kg}$

Required percentage = $(18 - 11.25)/18 \times 100$

= $6.75/18 \times 100$

= 37.5%

Hence, the answer is option C.

7. Questions

Let the number of employees who work in company Q = $190x$

Number of employees who work in company P = $220x$

Number of employees who work the morning shift of company P = $3/11 \times 220x = 60x$

Number of employees who work the night and morning shift together of company P = $5/11 \times 220x = 100x$

Number of employees who work the night shift of company P = $100x - 60x = 40x$

Number of employees who work in the evening shift of company Q = $190x \times 1/4 = 47.5x$

$47.5x + 40x = 210$

$x = 2.4$

Number of employees who work in company P = 528

Number of employees who work in morning shift in company P = 144

Number of employees who work in evening shift in company P = 288

Number of employees who work in the night shift of company P = 96

Number of employees who work in company Q = 456

@QuizByMickey

Number of employees who work in morning shift in company Q = 190 ($41.66\% = 5/12$)

Number of employees who work in evening shift in company Q = 114

Number of employees who work in night shift of company Q = 152

Number of employees who work in evening shift of company R = y

Total number of employees who work in company R = 7y

If $y = 114 + 34 = 148$, then $7y = 148 \times 7 = 1036$ (not possible)

If $y = 114 - 34 = 80$, then $7y = 560$

Number of employees who work in morning shift of company R = $2/7 \times 560 = 160$

Number of employees who work in night shift of company R = $6/7 \times 560 - 160 = 320$

Number of employees who work in evening shift of company R = $560 - 160 - 320 = 80$

Total number of employees in company S = $560/224 \times 250 = 625$

Number of employees who work in morning shift in company S = 28% of 625 = 175

Number of employees who work in evening shift in company S = 32% of 625 = 200

Number of employees who work in night shift of company S = $625 - 175 - 200 = 250$

Number of employees who work in evening shift of company T = 210

Total number of employees in company T = $210/7 \times 16 = 480$

Number of employees who work in morning shift in company T = 25% of 480 = 120

Number of employees who work in night shift of company T = $480 - 210 - 120 = 150$

Company	Total number of employees	Number of employees who work in morning shift	Number of employees who work in evening shift	Number of employees who work in night shift
P	528	144	288	96
Q	456	190	114	152
R	560	160	80	320
S	625	175	200	250
T	480	120	210	150

Answer: A

According to the question,

Difference between the number of employees who work in the evening and night shift in company Q = M = $152 - 114 = 38$

Difference between the number of employees who work in morning and night shift of company S = N = $250 - 175 = 75$

$$M + N = 113$$

Hence, the answer is option A.

8. Questions

Let the number of employees who work in company Q = $190x$

Number of employees who work in company P = $220x$

Number of employees who work the morning shift of company P = $\frac{3}{11} * 220x = 60x$

Number of employees who work the night and morning shift together of company P = $\frac{5}{11} * 220x = 100x$

Number of employees who work the night shift of company P = $100x - 60x = 40x$

Number of employees who work in the evening shift of company Q = $190x * \frac{1}{4} = 47.5x$

$$47.5x + 40x = 210$$

$$x = 2.4$$

Number of employees who work in company P = 528

Number of employees who work in morning shift in company P = 144

Number of employees who work in evening shift in company P = 288

Number of employees who work in the night shift of company P = 96

Number of employees who work in company Q = 456

Number of employees who work in morning shift in company Q = 190 (41.66% = $\frac{5}{12}$)

Number of employees who work in evening shift in company Q = 114

Number of employees who work in night shift of company Q = 152

Number of employees who work in evening shift of company R = y

Total number of employees who work in company R = 7y

If $y = 114 + 34 = 148$, then $7y = 148 * 7 = 1036$ (not possible)

If $y = 114 - 34 = 80$, then $7y = 560$

Number of employees who work in morning shift of company R = $\frac{2}{7} * 560 = 160$

Number of employees who work in night shift of company R = $\frac{6}{7} * 560 - 160 = 320$

Number of employees who work in evening shift of company R = $560 - 160 - 320 = 80$

Total number of employees in company S = $\frac{560}{224} * 250 = 625$

Number of employees who work in morning shift in company S = 28% of 625 = 175

Number of employees who work in evening shift in company S = 32% of 625 = 200

Number of employees who work in night shift of company S = $625 - 175 - 200 = 250$

Number of employees who work in evening shift of company T = 210

Total number of employees in company T = $210/7 \times 16 = 480$

Number of employees who work in morning shift in company T = 25% of 480 = 120

Number of employees who work in night shift of company T = $480 - 210 - 120 = 150$

Company	Total number of employees	Number of employees who work in morning shift	Number of employees who work in evening shift	Number of employees who work in night shift
P	528	144	288	96
Q	456	190	114	152
R	560	160	80	320
S	625	175	200	250
T	480	120	210	150

Answer: C

Quantity I: Average number of employees who work in morning shift in company R and T = $(160+120)/2 = 140$

Quantity II: Average number of employees who work in night shift in company Q and T = $(152+150)/2 = 151$

$140+15 > 151+2$

$155 > 153$

Hence, the answer is option C

9. Questions

Let the number of employees who work in company Q = $190x$

Number of employees who work in company P = $220x$

Number of employees who work the morning shift of company P = $3/11 \times 220x = 60x$

Number of employees who work the night and morning shift together of company P = $5/11 \times 220x = 100x$

Number of employees who work the night shift of company P = $100x - 60x = 40x$

Number of employees who work in the evening shift of company Q = $190x \times 1/4 = 47.5x$

$47.5x + 40x = 210$

$x = 2.4$

Number of employees who work in company P = 528

Number of employees who work in morning shift in company P = 144

Number of employees who work in evening shift in company P = 288

Number of employees who work in the night shift of company P = 96

Number of employees who work in company Q = 456

Number of employees who work in morning shift in company Q = 190 (41.66% = 5/12)

Number of employees who work in evening shift in company Q = 114

Number of employees who work in night shift of company Q = 152

Number of employees who work in evening shift of company R = y

Total number of employees who work in company R = 7y

If $y = 114 + 34 = 148$, then $7y = 148 \times 7 = 1036$ (not possible)

If $y = 114 - 34 = 80$, then $7y = 560$

Number of employees who work in morning shift of company R = $\frac{2}{7} \times 560 = 160$

Number of employees who work in night shift of company R = $\frac{6}{7} \times 560 - 160 = 320$

Number of employees who work in evening shift of company R = $560 - 160 - 320 = 80$

Total number of employees in company S = $\frac{560}{224} \times 250 = 625$

Number of employees who work in morning shift in company S = 28% of 625 = 175

Number of employees who work in evening shift in company S = 32% of 625 = 200

Number of employees who work in night shift of company S = $625 - 175 - 200 = 250$

Number of employees who work in evening shift of company T = 210

Total number of employees in company T = $\frac{210}{7} \times 16 = 480$

Number of employees who work in morning shift in company T = 25% of 480 = 120

Number of employees who work in night shift of company T = $480 - 210 - 120 = 150$

Company	Total number of employees	Number of employees who work in morning shift	Number of employees who work in evening shift	Number of employees who work in night shift
P	528	144	288	96
Q	456	190	114	152
R	560	160	80	320
S	625	175	200	250
T	480	120	210	150

Answer: D

Total number of employees in company Q = 456

Number of females = $\frac{456}{3} = 152$

Number of males = $152 \times 2 = 304$

Number of female employees on night shift = x

Number of male employees on night shift = $3x$

$$152/4 = x$$

$$x = 38$$

Number of females in night shift = 38

Number of females in morning and evening shift = $152 - 38 = 114$

Number of males in night and morning shift = $304 - 114 = 190$

$$190 - 114 = 76$$

Hence, the answer is option D.

10. Questions

Let the number of employees who work in company Q = $190x$

Number of employees who work in company P = $220x$

Number of employees who work the morning shift of company P = $3/11 * 220x = 60x$

Number of employees who work the night and morning shift together of company P = $5/11 * 220x = 100x$

Number of employees who work the night shift of company P = $100x - 60x = 40x$

Number of employees who work in the evening shift of company Q = $190x * 1/4 = 47.5x$

$$47.5x + 40x = 210$$

$$x = 2.4$$

Number of employees who work in company P = 528

Number of employees who work in morning shift in company P = 144

Number of employees who work in evening shift in company P = 288

Number of employees who work in the night shift of company P = 96

Number of employees who work in company Q = 456

Number of employees who work in morning shift in company Q = 190 (41.66% = $5/12$)

Number of employees who work in evening shift in company Q = 114

Number of employees who work in night shift of company Q = 152

Number of employees who work in evening shift of company R = y

Total number of employees who work in company R = $7y$

If $y = 114 + 34 = 148$, then $7y = 148 * 7 = 1036$ (not possible)

If $y = 114 - 34 = 80$, then $7y = 560$

Number of employees who work in morning shift of company R = $2/7 * 560 = 160$

Number of employees who work in night shift of company R = $6/7 * 560 - 160 = 320$

Number of employees who work in evening shift of company R = $560 - 160 - 320 = 80$

Total number of employees in company S = $560/224 \times 250 = 625$

Number of employees who work in morning shift in company S = 28% of $625 = 175$

Number of employees who work in evening shift in company S = 32% of $625 = 200$

Number of employees who work in night shift of company S = $625 - 175 - 200 = 250$

Number of employees who work in evening shift of company T = 210

Total number of employees in company T = $210/7 \times 16 = 480$

Number of employees who work in morning shift in company T = 25% of $480 = 120$

Number of employees who work in night shift of company T = $480 - 210 - 120 = 150$

Company	Total number of employees	Number of employees who work in morning shift	Number of employees who work in evening shift	Number of employees who work in night shift
P	528	144	288	96
Q	456	190	114	152
R	560	160	80	320
S	625	175	200	250
T	480	120	210	150

Answer: B

Number of employees in company A = $625/5 \times 4 = 500$

Number of employees working in evening shift in company A = 65% of $500 = 325$

Number of employees working in evening shift in company P = 288

$325 - 288 = 37$

Hence, the answer is option B.

11. Questions

Let the number of employees who work in company Q = $190x$ @QuizByMickey

Number of employees who work in company P = $220x$

Number of employees who work the morning shift of company P = $3/11 \times 220x = 60x$

Number of employees who work the night and morning shift together of company P = $5/11 \times 220x = 100x$

Number of employees who work the night shift of company P = $100x - 60x = 40x$

Number of employees who work in the evening shift of company Q = $190x \times 1/4 = 47.5x$

$47.5x + 40x = 210$

$x = 2.4$

Number of employees who work in company P = 528

Number of employees who work in morning shift in company P = 144

Number of employees who work in evening shift in company P = 288

Number of employees who work in the night shift of company P = 96

Number of employees who work in company Q = 456

Number of employees who work in morning shift in company Q = 190 (41.66% = 5/12)

Number of employees who work in evening shift in company Q = 114

Number of employees who work in night shift of company Q = 152

Number of employees who work in evening shift of company R = y

Total number of employees who work in company R = 7y

If $y = 114 + 34 = 148$, then $7y = 148 \times 7 = 1036$ (not possible)

If $y = 114 - 34 = 80$, then $7y = 560$

Number of employees who work in morning shift of company R = $\frac{2}{7} \times 560 = 160$

Number of employees who work in night shift of company R = $\frac{6}{7} \times 560 - 160 = 320$

Number of employees who work in evening shift of company R = $560 - 160 - 320 = 80$

Total number of employees in company S = $\frac{560}{224} \times 250 = 625$

Number of employees who work in morning shift in company S = 28% of 625 = 175

Number of employees who work in evening shift in company S = 32% of 625 = 200

Number of employees who work in night shift of company S = $625 - 175 - 200 = 250$

Number of employees who work in evening shift of company T = 210

Total number of employees in company T = $\frac{210}{7} \times 16 = 480$

Number of employees who work in morning shift in company T = 25% of 480 = 120

Number of employees who work in night shift of company T = $480 - 210 - 120 = 150$

Company	Total number of employees	Number of employees who work in morning shift	Number of employees who work in evening shift	Number of employees who work in night shift
P	528	144	288	96
Q	456	190	114	152
R	560	160	80	320
S	625	175	200	250
T	480	120	210	150

Answer: C

Average number of employees working in night shift in all companies = $(96+152+320+250+150)/5 = 193.6$

$$193.6/528 = 36.66\%$$

$$W = 20$$

Average number of employees in R and T together = 520

$$520 \times 20\%$$

$$= 104$$

Hence, the answer is option C.

12. Questions

Answer: D

$$x^2 - 40x + 384 = 0$$

$$x^2 - 24x - 16x + 384 = 0$$

$$x(x-24) - 16(x-24) = 0$$

$$x = 16 \text{ or } 24$$

$$A = 24 \text{ and } B = 16$$

$$I: [3B - 2], [2(A+B) + 14], [B^2 + 23], ?, [3(A-B)^3 + 141], [839 \times (B)^{0.5}]$$

$$46, 94, 279, ?, 1677, 3356$$

Pattern:

$$46 \quad 94 \quad 279 \quad 560 \quad 1677 \quad 3356$$

$$\times 2+2 \quad \times 3-3 \quad \times 2+2 \quad \times 3-3 \quad \times 2+2$$

Missing term = 560

$$II: 78, [6A + 15], ?, [9(A+2B)+24], 852, [(B-5)^3 - 38]$$

$$78, 159, ?, 528, 852, 1293$$

Pattern:

$$78 \quad 159 \quad 303 \quad 528 \quad 852 \quad 1293$$

$$+9^2 \quad +12^2 \quad +15^2 \quad +18^2 \quad +21^2$$

Missing term = 303

$$\text{Required difference} = 560 - 303 = 257$$

Hence, the answer is option D.

13. Questions

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Answer: E

Series: 12, 21, 30, 43, 60, 79

Pattern:

12	19	30	43	60	79
	+7	+11	+13	+17	+19

Wrong term = 21

Correct term = P = 19

$7P = 7 * 19 = 133$

New series:

133	140	151	164	181	200
	+7	+11	+13	+17	+19

Y = 200

Thus, percentage = $(2 * 19/200) * 100 = 19\%$

Hence, the answer is option E.

14. Questions

Answer: C

I: 336, 348, 364, 380, 408, 436

Pattern:

336	348	364	384	408	436
	+12	+16	+20	+24	+28

Wrong term = X = 380

II: 421, 457, 506, 580, 690, 849

Pattern:

421	457	506	580	690	849
+33	+49	+74	+110	+159	
	$+4^2$	$+5^2$	$+6^2$	$+7^2$	

Wrong term = Y = 421

III: 56, 364, 1820, 6370, 12740, 6375

Pattern:

56	364	1820	6370	12740	6375
*6.5	*5	*3.5	*2		*0.5

Wrong term =Z= 6375

$$[(Z-Y)/2] \div [(25\% \text{ of } X)/5+7]$$

$$((6375-421)/2) \div ((25\% \text{ of } 380)/5+7)$$

$$2977 \div 26$$

$$= 114.5$$

Hence, the answer is option C.

15. Questions

Answer: A

I: 103, 128, 249, 538, (P+Q), 2028

Pattern:

103	128	249	538	1067	2028
$+5^2$	$+11^2$	$+17^2$	$+23^2$	$+31^2$	

$$P+Q = 1067$$

II: P, R, 436, 315, 484, 259, 548

Pattern:

404	355	436	315	484	259	548
-7^2	$+9^2$	-11^2	$+13^2$	-15^2	$+17^2$	

$$P = 404$$

$$R = 355$$

$$Q = 1067 - 404 = 663$$

$$1.8R = 355 \times 1.8 = 639$$

$$1.8R < Q$$

Hence, the answer is option A.

16. Questions

Answer: C

$$\text{Amount} = P * [1 + R/100]^n$$

$$SI = PNR/100$$

$$\text{Borrowed amount} = \text{Rs } (12A + 10680)$$

$$\text{let } 12A + 10680 = 512x$$

$$\text{Amount after three years} = 512x * [112.5/100]^3 = 729x$$

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According to question

$$\text{Interest earned from Shruti} = 512x \times \frac{3}{5} \times 45\% = 138.24x$$

$$\text{Interest earned from Prachi} = 512x \times \frac{2}{5} \times \frac{61}{64} = 195.2x$$

$$\text{Interest given to bank} = 729x - 512x = 217x$$

$$\text{Profit} = 333.44x - 217x = 10479.6$$

$$116.44x = 10479.6$$

$$x = 90$$

$$12A + 10680 = 512 \times 90$$

$$12A = 35400$$

$$A = 2950$$

(I) A is not divisible by 11

$$(II) 2950 \times 24\% = 3648 - 2950$$

$$708 = 698 \text{ false}$$

$$(III) 2950 + 75 = 3025 = 55^2 \text{ true}$$

Hence the answer is option C.

17. Questions

Answer: B

Difference between downstream speed of P and upstream speed of R = 9 kmph

Downstream distance covered by boat P in (Z) hours is 60% more than the upstream distance covered by boat R in same time

Let downstream speed of boat P = 8a

Upstream speed of boat R = 5a

$$3a = 9$$

$$a = 3$$

Downstream speed of boat P = 24 kmph

Upstream speed of boat R = 15 kmph

speed of river = (X-7) kmph

Boat P can cover 156 km upstream distance in (X) hours

Upstream speed of boat P = 156/X

$$156/X = 24 - 2(X-7)$$

$$78 = X(12 - X + 7)$$

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$$78 = 19X - X^2$$

$$X^2 - 19X + 78 = 0$$

$$X = 13 \text{ or } 6$$

X can not be 6 so $X = 13$

Boat Q covers Y km downstream distance in 8 hours

Only the Value of X can be determined

Hence, the answer is option B.

18. Questions

Answer: E

The age of C after 6 years is 64 (perfect square and perfect cube).

Present age of C = 58 years

Ratio of present age of A and age of B after 9 years = 7:6

Age of A > age of B

Let present age of A = $7x$ years

Present age of B = $6x - 9$ years

Present age of D = $7x - 8$ years

$$(7x + 58 - 12) = 8/7 (6x - 9 + 7x - 8 + 16)$$

$$49x + 322 = 104x - 8$$

$$55x = 330$$

$$x = 6$$

The sum of the present age of B and C = $58 + 6x - 9 = 58 + 27 = 85$ years

Hence, the answer is option E.

19. Questions

Answer: B

According to the questions

Let the number be ABC

On reversing digits, the number becomes 297 less; it means $C < A$, and after reversing the digits, when we subtract A from C, the resultant is 7.

C can be any number from 1-6

If C is 2, then A will be 5

$A = B + C$ so $B = 3$ and number = 532

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$$532-235 = 297$$

If C is 3, then A will be 6, and B will also be 3, which is not possible

If C is 4, then A will be 7, and B will be 3

Number = 734

$$734-437 = 297$$

If C is 5, then A will be 8 and B = 3

Number = 835

$$835 - 538 = 297$$

If C is 6, then A will be 9 and B = 3

Number = 936

Only the digit at the tens place can be determined

Hence, the answer is option B.

20. Questions

Answer: C

Time taken by M to finish the work = $(Y - 4)$ days

Time taken by O to finish the work = $(Y + 5)$ days

Efficiency of N and O = 6:7

Let the time taken by N to finish the work = $7x$ days

Time taken by O to finish the work = $6x$ days

$$\text{So, } 6x = Y + 5$$

$$Y = 6x - 5$$

Time taken by M to finish the work = $6x - 5 - 4 = (6x - 9)$ days

let total work = $42x$ units

Efficiency of M = $14x / (2x - 3)$

Efficiency of N = 6

$$[(14x / (2x - 3)) + 6] * 105 / 8 = 42x$$

$$16x^2 - 89x + 45 = 0$$

$$x = 80/16 \text{ or } 9/16$$

$$x = 5$$

Time taken by O to finish the work = 30 days

$$30 = Y + 5$$

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$Y = 25$

Option C : $(25)^2 = 625 = 6+2+5 = 13$ (true)

Hence, the answer is option C.

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